

Answer ALL questions.

Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☐. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☐.

- 1** Brazil is the largest producer of sugarcane.

The photograph shows sugarcane plants growing in Brazil.



(Source: © jeep2499/Shutterstock)

- (a) Products from sugarcane, such as starch and cellulose, are used to make bioplastic.

- (i) How many of the following statements about **starch** are correct?

- contains α -glucose and β -glucose molecules
- contains amylose and amylopectin
- stored in the tonoplast

(1)

☐ **A** none

☐ **B** one

☐ **C** two

☐ **D** three



(ii) How many of the following statements about a **cellulose** molecule are correct?

- contains α -glucose and β -glucose molecules
- contains 1,4 glycosidic bonds
- forms hydrogen bonds with other cellulose molecules

(1)

- ☐ **A** none
- ☐ **B** one
- ☐ **C** two
- ☐ **D** three

(b) Bioplastic bags made from sugarcane take up to six months to decompose.

Bacteria secrete enzymes onto the bags during the process of decomposition.

(i) Name **two** structures found in bacterial cells that are involved in the synthesis of these enzymes.

(1)

1

2



- (ii) Decomposition of bioplastic bags occurs faster if there is increased bacterial growth.

Explain the conditions needed for increased bacterial growth.

(3)

(Total for Question 1 = 6 marks)

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